DATA ANALYTICS • STATISTICS

What is a Confidence Interval?

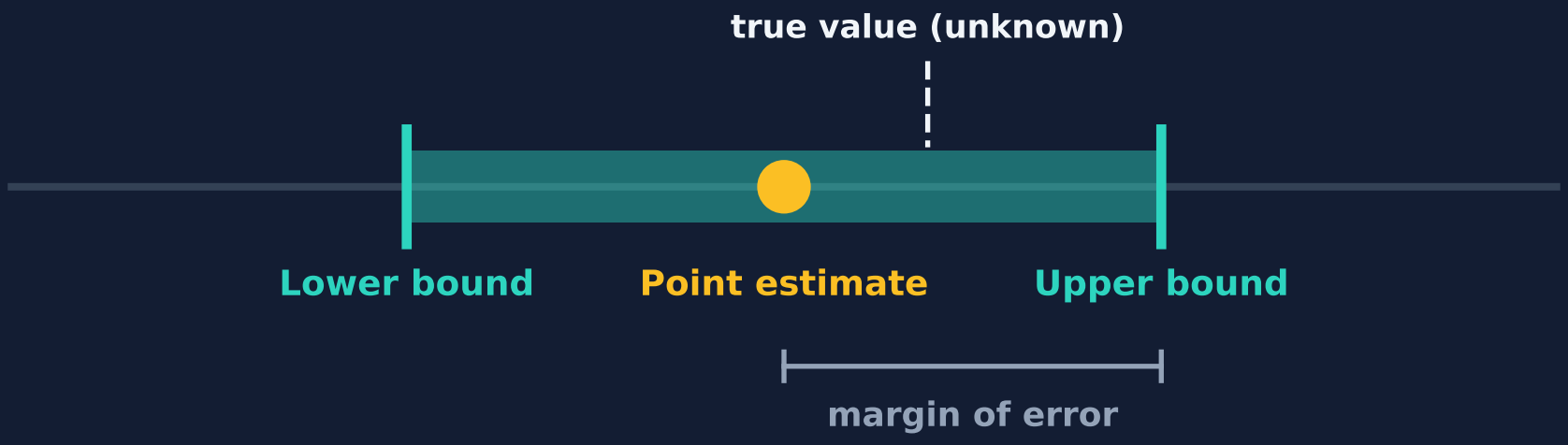
IN ONE LINE

A range of values, calculated from a sample, that is likely to contain the true population value.

It turns a single estimate into an estimate with its uncertainty.

PICTURE IT

Sample data gives us one estimate. The truth lies somewhere around it.



Point estimate

Your sample mean or proportion

Margin of error

How far the estimate can reasonably miss

Confidence level

Usually 90%, 95% or 99%

THE FORMULA

$$CI = \bar{x} \pm z \times \frac{s}{\sqrt{n}}$$

$\bar{x}$ = sample mean z = 1.96 for 95%

s = std. deviation n = sample size

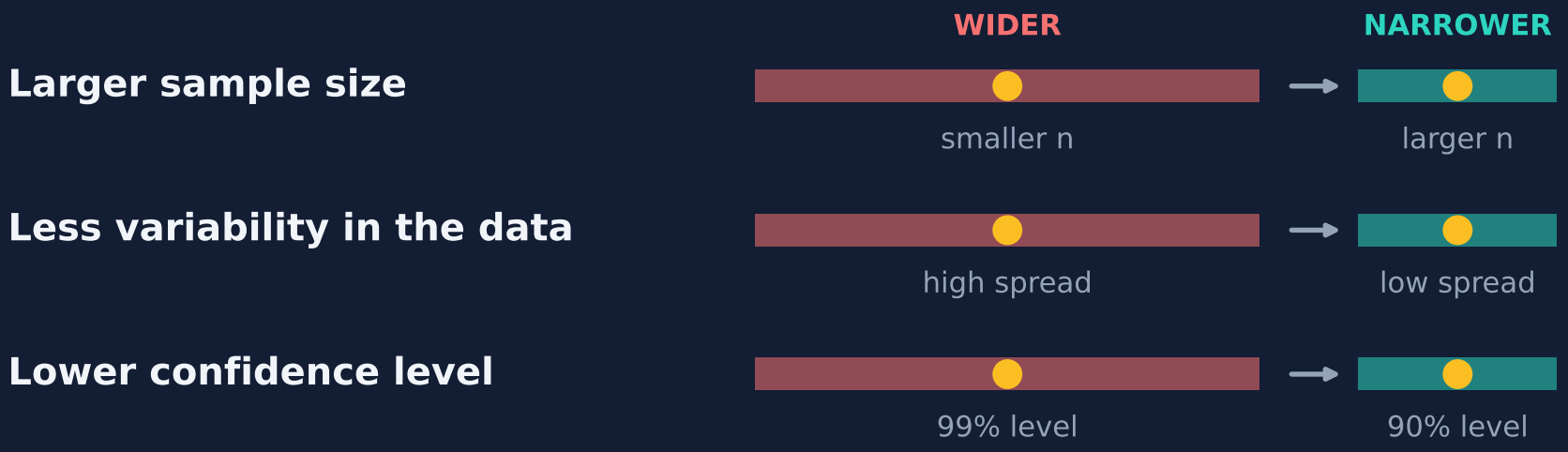
Swipe to see why it matters →

Why It Matters & How to Read It

WHY WE USE IT

- **We rarely see the whole population.**
  We estimate from samples, and samples vary.
- **A single number hides uncertainty.**
  The interval shows how precise the estimate is.
- **It supports better decisions.**
  Narrow means more precise, wide means be careful.

WHAT CHANGES THE WIDTH?



A COMMON MISTAKE

✗ Wrong

“There is a 95% chance the true mean is inside this interval.”

✓ Better

If we repeated the sampling many times, about 95% of the intervals built this way would capture the true mean.

Swipe for real-world applications →

WHERE IT IS USED

Real-World Applications

Healthcare

Estimating how well a treatment works in clinical trials

Business

A/B tests: is a new design really better, or just noise?

Surveys & Polls

The “±3%” margin of error in election and market polls

Manufacturing

Quality control: is average product size within spec?

Environment

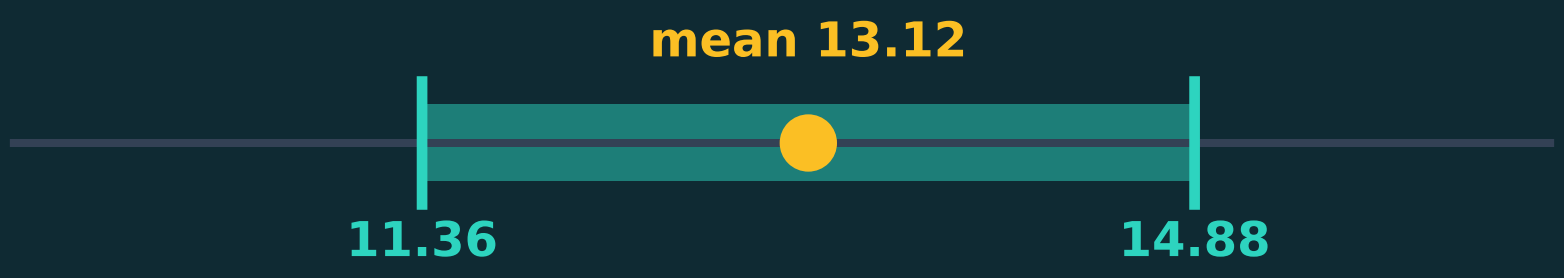
Comparing air quality or pollution levels across regions

Finance

Ranges around forecasts, returns and risk estimates

MY PROJECT: AIR QUALITY INDEX (AQI)

Highest mean AQI among six US states: California



95% confidence interval, margin of error ≈ ±1.76

KEY TAKEAWAY

A confidence interval shows how precisely a sample estimates the truth.

Always read it in the context of the dataset and its limitations.

Python • Pandas • Jupyter | github.com/ankitt994/AQI-Confidence-Interval-Analysis